

Stoquify POS and Sales-to-Cash Contract and Control Freeze

Status: **DRAFT — REQUIRES EXPERT REVIEW**

Program: `STOQUIFY-POS-SALES-TO-CASH`

Roadmap authority: `STOQUIFY_POS_AND_SALES_TO_CASH_MASTER_IMPLEMENTATION_ROADMAP_2026-08-17.md`

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This is the M0/M1 implementation contract. It authorizes design and read-only verification, not product mutation, provider activation, fiscal claims or production release. All statutory and accounting conclusions remain subject to dated qualified-human review.

1. Program boundary

In scope

- Immediate POS checkout and delivery/on-account order-to-cash.
- One sale finalizer: `commitPOSSale`.
- Shared payment, receipt/fiscal-source, inventory/valuation, accounting and reconciliation kernels.
- Cashier sessions, drawers, governed business days, statements, reconciliation and close.
- Returns, refunds, credit notes, stock corrections and journal reversals as linked compensating facts.
- Tenant, organization, location, terminal and actor access boundaries.

Out of scope until a later approved gate

- A second sale-finalization service.
- Loyalty, CRM campaigns, recommendation engines, generic analytics and broad UI redesign.
- Unproven multi-currency, inclusive tax, UOM, lot, serial, expiry, offline electronic capture or fiscal certification.
- National tax or receipt rules hardcoded into shared code.

2. Authoritative owners

Fact or transition	Sole authoritative owner	Reused by
Immediate POS financial completion	<code>commitPOSSale</code> transaction boundary	POS workstation and retry/result registry
Electronic provider truth	Named provider adapter plus provider-event ingestion	POS, AR collections and refunds
Payment settlement truth	Statement ingestion and reconciliation kernel	POS close, AR, treasury and accounting close
Receipt/fiscal source	Immutable receipt/fiscal-source materializer	POS and invoicing

Fact or transition	Sole authoritative owner	Reused by
Receipt delivery	Durable delivery worker	Print, email, messaging and public-token delivery
Physical stock consequence	Existing stock-event and valuation kernel	POS goods issue, fulfillment, returns and corrections
Journal posting	Existing ledger posting kernel	POS, invoices, settlements, fees and corrections
Reservation	Reservation aggregate	Sales order and fulfillment
Delivery/on-account orchestration	SalesOrder/Fulfillment/Invoice aggregates	O2C workbench
Business-day close evidence	Governed business-day and statement services	Retail operations, treasury and accounting close

No adapter, route, worker or UI may own a second copy of these facts.

3. Universal command contract

Every material write command must carry or derive the following immutable context:

```
commandId/clientCommitId
commandType + commandVersion
organizationId + locationId
terminalId/deviceId when applicable
actorId + authenticated session id
authorization snapshot: permission, module entitlement, assignment
fresh-auth evidence and approval id when policy requires them
expected aggregate version or explicit create-only precondition
canonical request payload hash
business date + UTC occurred-at/received-at timestamps
currency + exact monetary values
correlationId + causationId
country-pack/provider/policy versions when applicable
```

Rules:

1. The first accepted `(organizationId, commandType, commandId)` stores the canonical payload hash and immutable result envelope.
2. A retry with the same payload returns the original result without repeating stock, payment, receipt, fiscal or ledger consequences.
3. The same key with a different payload is rejected as a conflict and audited.
4. Scope, entitlement, permission, assignment, fresh authentication and approval are checked inside the authoritative command boundary, not only in the UI.
5. Timeouts and provider uncertainty remain pending/unknown; they are never translated into success.

4. State contracts

The names below are semantic contracts. Implementations may map existing enums, but may not collapse materially different states.

Aggregate	Required minimum states	Forbidden shortcut
POS cart	OPEN , CHECKOUT_PENDING , COMPLETED , ABANDONED , CONFLICT	Mutating a completed cart
POS sale	COMMITTING , COMPLETED , CORRECTION_REQUIRED	Deleting or rewriting a completed sale
Payment attempt	CREATED , PENDING_PROVIDER , UNKNOWN , CAPTURED , FAILED , REVERSED , REFUNDED	Treating timeout/unknown as captured
Settlement item	UNMATCHED , MATCHED , SUSPENSE , RESOLVED	Manufacturing a match without evidence
Receipt source	MATERIALIZED , FISCAL_PENDING , FISCAL_ACCEPTED , FISCAL_REJECTED , CORRECTED	Equating delivery with existence
Receipt delivery	QUEUED , DELIVERED , RETRY_REQUIRED , DEAD_LETTER	Rolling back a sale when delivery fails
Cashier session	OPEN , DECLARATION_PENDING , DECLARED , REVIEWED , CLOSED	Closing solely because a cashier logs out
Business day	OPEN , CLOSE_PENDING , CLOSED , INVALIDATED , REOPEN_REQUIRES_APPROVAL	Hiding late facts after close
Sales order	DRAFT , CONFIRMED , PARTIALLY_FULFILLED , FULFILLED , CANCELLED	Posting revenue/COGS on confirmation
Reservation	ACTIVE , PARTIALLY_CONSUMED , CONSUMED , RELEASED , EXPIRED	Reducing on-hand or posting COGS
Fulfillment	RELEASED , PICKED , PACKED , ISSUED , DELIVERED , REVERSED	Issuing stock twice
Invoice	DRAFT , ISSUED , PARTIALLY_PAID , PAID , CREDITED , VOIDED_BY_CORRECTION	Billing ineligible quantity
Return	REQUESTED , AUTHORIZED , RECEIVED , INSPECTED , DISPOSITIONED , COMPLETED	Increasing sellable stock before disposition

All transitions require explicit allowed-from states, version preconditions, owner, evidence, audit fields, failure semantics and compensating transition.

5. Money, quantity, tax and time

- The repository indicates Cameroon/XAF/EN-FR as a development candidate only. It is not a production or legal conclusion.
- Persist monetary values exactly; do not use binary floating point for financial calculations.
- A qualified country-pack reviewer must approve currency minor units, rounding order, tax inclusion/exclusion, tax point, receipt numbering, correction and retention rules.
- Persist quantity and valuation precision explicitly. Unsupported UOM, lot, serial, expiry, negative-stock or multi-location combinations fail closed.
- Store event time in UTC and preserve the source/provider time. Derive business date from the versioned organization/location timezone and business-day policy; never equate business date with server date.

6. Accounting and inventory event matrix

Event	Quantity	Reserved	Journal consequence
POS sale completion and physical hand-over	decrease on-hand	none	Dr cash/provider clearing; Cr revenue/tax; Dr COGS; Cr inventory
Electronic authorization pending/unknown	none	none	no successful-payment posting
Provider capture confirmed	none	none	provider-clearing consequence exactly once under approved policy
Settlement statement match	none	none	Dr bank/fees; Cr provider clearing, with suspense for mismatch
Sales-order confirmation	none	none	none
Reservation create/release	none	increase/decrease	none
Physical goods issue	decrease on-hand	consume	Dr COGS; Cr inventory exactly once
Eligible invoice issue	none	none	Dr AR; Cr revenue/tax exactly once
Customer payment allocation	none	none	Dr cash/provider clearing; Cr AR
Return disposition to sellable	increase on-hand	none	linked COGS/inventory correction as approved
Damage/write-off	decrease or no sellable increase	none	linked approved inventory loss posting

The exact accounts and tax/fiscal semantics are configuration/country-pack responsibilities and require controller review. Every posting line links the immutable operational source, command result and correction lineage.

7. Access and approval matrix

Command class	Permission + entitlement	Assignment/scope	Fresh auth	Maker-checker
Ordinary cash sale	required	organization, location, active terminal/session	policy-based	no unless threshold/risk rule
Electronic capture/refund	required	organization, location, provider account	required for refund/high risk	required above policy threshold
Void/return/credit note	required	source sale/order and allowed location	required	required when material; self-approval forbidden
Drawer declaration/business-day close	required	claimed terminal/location/business day	required	reviewer separate from declarer where required
Stock issue/reversal/write-off	required	inventory location	required for reversal/write-off	required above threshold
Country/provider/capability activation	administrative permission	organization/package/ringing	required	two-person approval

All denials are auditable and redacted. IDs supplied by clients are re-scoped server-side; a valid identifier from another tenant/location must not disclose existence.

8. Receipt, fiscal and delivery contract

1. Completion creates immutable receipt/fiscal-source evidence in the transaction or through a transactional outbox tied to it.
2. Receipt/fiscal numbering is owned by a versioned country pack or authority adapter. Offline replay cannot allocate final fiscal numbers unless that pack explicitly authorizes it.
3. Print/email/message delivery is an independent retrievable concern. Delivery failure leaves the sale visible and complete with `RETRY_REQUIRED` evidence.
4. Correction creates linked corrective source evidence; it never mutates the original receipt, invoice or fiscal fact.
5. Public delivery tokens are scoped, expiring, revocable, non-enumerable and audited without exposing unnecessary PII.

9. Provider and reconciliation contract

- Provider requests use stable idempotency keys and persist a local attempt before external submission.
- Callbacks require signature verification, replay protection, provider account/currency/amount validation and tenant/location resolution from trusted configuration.
- Provider events are append-only. Out-of-order and duplicate events are retained and deterministically reduced.
- Captured, settled, fee, reversal, refund, dispute and chargeback are separate semantics.
- Bank/provider statement ingestion is source-hashed and replay-safe. Unmatched or contradictory evidence enters owned suspense; it never becomes an invented success.

10. Offline and device contract

- Offline events are device-signed, monotonically ordered, expiration-bounded and bound to organization, location, terminal, actor and policy version.
- The server verifies signature, revocation, sequence, payload hash, price/tax snapshot and capability eligibility before replay.
- Duplicate replay returns the original result; sequence gaps and payload conflicts become operator-resolvable cases.
- Cash-only offline is the recommended first scope. Electronic, final fiscal numbering and unbounded catalog/price operation remain disabled until separately certified.

11. Data change and migration policy

- Additive, deploy-safe migrations only; no reset, destructive repair or mutation/deletion of completed sales, payments, stock movements, receipts, fiscal documents, journals or audit records.
- Constraints become authoritative only after source profiling, deterministic backfill, quarantine of contradictions, tie-out and reconciliation evidence.
- Financial and stock corrections use source-linked compensating records.
- Every migration slice includes forward SQL, compatibility window, backfill, verification query, rollback/disable plan and real PostgreSQL evidence.

12. Evidence and privacy contract

Every gate artifact identifies candidate commit/tree, dirty-state hash, command, environment, configuration/country/provider versions, start/end time, exit code, result, source hashes, owner and limitations. Local mocks and unit tests are labeled local evidence; they are never presented as production proof.

Never store secrets, tokens, webhook signatures, PAN, CVV, PIN, raw payment credentials or unnecessary customer PII in logs, fixtures, screenshots or evidence. Use stable redacted references and aggregates.

13. Gate disposition

Gate	Current disposition	What changes it
G0 Verification Foundation	BLOCKED	Resolve the 13 exact-hash migration findings, then complete the baseline ladder, real PostgreSQL and authenticated EN/FR harness proof, plus named pilot matrix
G1 Contract and Control Freeze	REQUIRES_EXPERT_REVIEW	Accountable approval of D-01 through D-11 and this versioned contract
G2–G9	BLOCKED_DEPENDENCY	Prior gates pass with immutable evidence

The first authorized product slice after G0/G1 is M2 client-commit result replay with PostgreSQL concurrency evidence. Until then, work is limited to verification, harness construction, evidence capture and approval preparation.